Platzhalter

Lo
go

Term paper

Sample title for a report

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Acronyms

CNC Computerized Numerical Control

SPS Speicherprogrammierbare Steuerung

1. Introduction

Project: Number Plate Detection using Jetson Nano
Edge deployment on NVIDIA Jetson Nano using YOLOv8-N (Nano) and YOLOv8-L (Large) models; local SQL database logging and dwell-time based billing.

1.1. Executive Summary

This project proposes a **single-camera license plate recognition (LPR) and billing system** for German airport and security gate scenarios. It combines:

- **YOLOv8 models** for fast, robust plate detection [**jocher2023ultralytics**].
- **OCR engines** such as EasyOCR/Tesseract for plate text extraction.
- A lightweight **SQLite database** for local logging, dwell-time calculation, and billing.
- Built-in handling of **exceptions** (e.g., police, emergency, diplomatic vehicles).

The system emphasizes real-time performance, scalability, and GDPR-compliance by performing all processing on edge hardware (Jetson Nano) without reliance on cloud servers.

1.2. Motivation and Relevance

European airports face high throughput and strict security/privacy requirements:

- Manual verification is slow and costly.
- Proprietary ANPR cameras are expensive and vendor-locked.
- YOLO-based edge systems offer flexibility, cost savings, and fast local decisions.

1.3. Key Objectives

1. Develop a real-time YOLOv8-Nano detection pipeline for German license plates.
2. Integrate OCR for text extraction, including suffixes (E, H) and color-coded diplomatic/military plates.
3. Log events in a GDPR-compliant local SQLite database.
4. Implement billing logic with dwell-time calculation and exemptions for special plates.
5. Validate under varied conditions (lighting, weather, skewed angles).

1.4. Contribution

This chapter establishes:

- The **technical motivation** for YOLO-based edge ANPR systems.
- A structured overview of the system's pipeline (detection → OCR → database → billing).
- The legal and operational relevance of deploying ANPR systems in German contexts.

1.5. Structure of the Report

The remainder of this report is structured as follows:

- **Application Domain Knowledge:** Practical use case for gesture/remote systems.
- **Hardware:** Arducam Pico4ML-BLE and Jetson Nano setup.
- **Algorithm:** CNN and YOLOv8 details.
- **KDD for TinyML:** Knowledge discovery methodology.
- **Development to Deployment:** Pipeline implementation, database, data preparation.
- **Verification and Conclusion:** Evaluation metrics and future scope.

Part I.

Domain Knowledge - Application

2. Application Domain Knowledge

2.1. Part Domain Knowledge — Application

Context. We deploy *Automatic Number Plate Recognition (ANPR)* on an embedded edge device (NVIDIA Jetson Nano B01 + USB UVC camera) to detect plates with a real-time object detector and read text via OCR. Typical use cases include barrierless parking, campus access control, and vehicle audit trails where on-device inference reduces latency and preserves data sovereignty [**nvidiaJetsonNano**; **anprSurvey**; **yolov5**; **easyocr**; **edgePrivacy**].

Table 2.1.: Domain checklist for edge ANPR (what matters and why)

Dimension	**What it means for our project**
Application	Barrierless entry/exit and logging on a closed site; on-device inference: low latency, no WAN dependence, and lower privacy risk [**edgePrivacy**].
Problem	Detect EU/German plates in unconstrained scenes (day/night, rain, glare), crop plates robustly, OCR the text, and attach policy (e.g., whitelist/exception) in near real time.
Data acquisition	We use a UVC USB camera (Microsoft LifeCam Show) mounted at 2.0 m–3.0 m height; focal direction tuned to capture rear/front plates at 3 m–8 m. Optional synthetic augmentation fills rare cases [**dataAug**].
Data quantity	For robust training/fine-tuning: ∼5–10k annotated plate instances for detection; ∼100–300 samples per frequent character trigram for OCR calibration; continuous logging for evaluation.
Data quality	Sharp plates with plate height $\geq$ 40–50 px in the crop, minimal motion blur (<1 px PSF), correct exposure ($\leq$1% saturated pixels in plate region).
Data relevance	German/EU plate layout (DE/EU bands, DIN 1451/FE-Schrift) and local fonts dominate; include neighboring countries' layouts for transferability [**anprSurvey**; **ocrSurvey**].
Outliers	Novel vanity formats, dirty/damaged plates, temporary paper plates, tow hooks occluding characters, non-plate rectangular signs triggering false positives.
Anomalies	Spoofed/fake plates, adversarial stickers, reflections producing duplicated contours, multi-plate scenes (car + bike rack), frames with no vehicle but high-contrast text.

2.1.1. Problem

Given a live RGB stream, output tuples ⟨timestamp, plate_id, bbox, confidences, plate_text, policy⟩ with end-to-end latency $\leq$ 200 ms at $\geq$ 15 fps. The system must be robust to weather (rain, snow), illumination (night, headlight glare), and pose (yaw/pitch) typical for driveway/parking entrances [**anprSurvey**].

2.1.2. Data Acquisition (Camera and Setup)

Camera: Microsoft LifeCam Show, UVC USB 2.0, 2 MP sensor. Operated at 640×480 or 1280×720 for stable fps on Nano. **Mounting:** Rigid pole/wall, angle 10°–25° downward; avoid facing bright road signs. **Exposure:** Lock exposure/white balance where possible to reduce inter-frame variance; enable short shutter and higher gain at night to reduce motion blur. **Ingestion:** V4L2/GStreamer to NVMM (zero-copy) where possible to keep GPU busy [**nvidiaJetsonNano**].

2.1.3. Data Quantity

- **Detection:** At least 5k diverse plate instances for fine-tuning (angles, distances, weather, day/night).
- **OCR:** Balancing samples across characters and common bigrams/trigrams (e.g., `AA`, `ER`, `MÜ`) to reduce bias [**ocrSurvey**].
- **Evaluation:** 10–20% held-out sequences per condition (day, night, rain) and at least 1 week of continuous logs for drift checks.

2.1.4. Data Quality

Quality gates before training/inference:

1. **Sharpness**: Laplacian variance > threshold; reject blurred frames.
2. **Brightness**: Mean plate ROI within [0.2, 0.8] (normalized); cap highlights.
3. **Plate Size**: Minimum plate height in frame and in crop; upscale only within 1.5× to limit artifacts.
4. **Label Consistency**: IoU agreement >0.5 across annotators for detection; OCR string validator (regex for country format).

2.1.5. Data Relevance (Formats & Policies)

German/EU plates follow regulated layouts (country band, district code, letters–digits, inspection badge positions). We encode format-aware validators and country-specific regex to improve OCR post-processing and reduce false accepts [**anprSurvey**; **euPlateStd**].

2.1.6. Outliers and Anomalies Handling

Table 2.2.: Operational outliers/anomalies and mitigations

Case	Mitigation
Dirty/damaged plates	Multi-crop super-resolution fallback; OCR with CTC and edit-distance voting; allow 1–2 char wildcard for whitelist hit.
Headlight glare/night	HDR exposure bracket or temporal fusion; polarized shield; train with synthetic flare/glare.
Non-plate text	Two-stage filtering: YOLO plate class → geometry/AR check → OCR; reject crops with invalid layout.
Fake/spoofed plates	Cross-check checksum-like constraints and blacklist patterns; temporal consistency across frames; optional backend verification.
Multiple plates	Track-by-detection; associate crops to tracks; per-track OCR majority vote to stabilize.

2.1.7. Data Sources & Licensing

Table 2.3.: Typical data sources and legal considerations (fill concrete sources you use)

Source	Use	Notes
Public ANPR datasets	Detector/OCR benchmarking	Check license; ensure EU plate coverage [**anprDataset**].
Own site recordings	Fine-tuning, domain fit	Obtain signage/consent; keep retention short; hash plates at rest [**gdpr**].
Synthetic generation	Rare cases, augmentations	Vary font/spacing, weather, motion blur; helpful for low-resource layouts [**dataAug**].

2.1.8. Edge Inference Pipeline (TikZ)

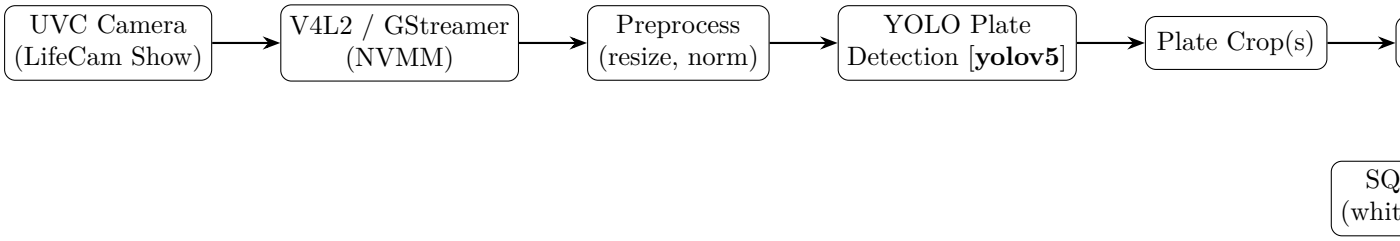

Figure 2.1.: On-device ANPR pipeline on Jetson Nano.

2.1.9. Privacy and Policy Notes (GDPR-aware)

Perform on-device inference; store only hashed plate strings plus salted last-4 preview where policy allows; enforce minimal retention and role-based access; display site signage and document purpose/legitimate interest [**gdpr**; **edgePrivacy**].

2.1.10. Operational KPIs

- Detection AP_{50} on held-out day/night; OCR CER/WER on plate crops.
- End-to-end latency (camera → decision), fps stability over 30 min.
- False accept / false reject on whitelist; drift rate vs. weather/time-of-day.

Part II.

Domain Knowledge - Hardware

3. Hardware XY

4. Sensor XY

Introduction

WS:cite books
applications, l

4.1. General

General description
cite books

4.2. Specific Sensor/Actor

cite board

4.3. Specification

- Cite the data sheet
- Extract te information from the data sheet
- Circuit Diagram

4.4. Library

4.4.1. Description

4.4.2. Installation

- data types
- data structure
- data quantity
- data quality

4.4.3. Functions

4.5. Example

In this section, a simple example is described in detail, which demonstrates how the library supports the sensor/actor.

4.5.1. Manual

4.5.2. Inside the Example

4.5.3. Code

4.5.4. Files

4.6. Calibration

cite method

4.7. Simple Code

4.8. Simple Application

4.9. Tests

4.9.1. Simple Function Test

4.9.2. Test all Functions

4.10. Further Readings

5. Actor XY

Introduction

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applications, l

5.1. General

General description
cite books

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- Cite the data sheet
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5.7. Simple Code

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5.9. Tests

5.9.1. Simple Function Test

5.9.2. Test all Functions

5.10. Further Readings

Part III.

Domain Knowledge - Machine Learning/Algorithms

6. Algorithm XY

Part IV.

Domain Knowledge - Tools

7. Python Tool XY

8. Hardware Tool XY

Part V.

Methodology

9. KDD XY

Part VI.

Developmenmt

10. Flow Chart Development XY

11. Database XY

12. Data Selection XY

13. Data Transformation XY

14. Data Mining XY

Part VII.

Development to Deployment

15. Machine Learning Pipeline XY

16. Database XY

Part VIII.

Deployment

17. Flow Chart Deployment XY

Part IX.

Monitoring

18. Monitoring XY

Part X.

Verification - Evaluation - Conclusion

19. Evaluation/Verification XY

20. To DoX Y

21. Conclusion XY

Part XI.

Appendix

22. Bill of Material

23. SBOM

requirements.txt

24. Package Example

24.1. Introduction

24.2. Description

24.3. Installation

```
pip packageExample
```

24.4. Example - Manual

24.5. Example

24.6. Example - Code

24.7. Example - Files

PackageExample.py

24.8. Further Reading

References

[Aba+16] M. Abadi et al. "TensorFlow: A System for Large-Scale Machine Learning". In: *12th USENIX Symposium on Operating Systems Design and Implementation (OSDI 16)*. Savannah, GA: USENIX Association, Nov. 2016, pp. 265–283. ISBN: 978-1-931971-33-1. URL: https://www.usenix.org/conference/osdi16/technical-sessions/presentation/abadi.

Part XII.

Hints - Examples for LaTeX - Read and Use

25. Hints

Please, use **`biber.exe`**.

If you want to create a symbol directory, call makeindex:
`makeindex %.nlo -s nomencl.ist`

25.1. Allgemeine mathematische Beschreibung Bézier-Kurve

Bézier curves are formed with the help of Bernstein polynomials. If at least two points and two tangents are known, control points can be determined with which a control polygon is formed. The course of the curve is oriented to this control polygon. The number of control points depends on the degree of the Bézier curve. A Bézier curve of degree n has n+1 control points. The Bézier curve is calculated using the De Casteljau algorithm.

Definition. In the interval [0; 1] the **Bernstein polynomial of nth degree** is defined by:

$$b_{i,n}(\lambda) = \binom{n}{i}(1-\lambda)^{n-i}\lambda^i, \quad \lambda \in [0,1], \quad i = 0, \ldots, n \tag{25.1}$$

Definition. The binomial coefficient is defined by:

$$\binom{n}{i} = \frac{n!}{i!(n-i)!}, \quad i = 0, \ldots, n \tag{25.2}$$

Here the Bernstein polynomials $b_{i,n}$ form a basis of the vector space $\mathbb{P}^n(I)$ of polynomials of degree at most n over I. Thus every polynomial of degree at most n can be written uniquely as a linear combination. [Far02]

Beispiel. Determination of Bernstein polynomials for $n = 3$ holds:

$$\begin{aligned}
b_{3,0}(\lambda) &= \binom{3}{0}(1-\lambda)^{3-0}\lambda^0 &&= (1-\lambda)^3 \\
b_{3,1}(\lambda) &= \binom{3}{1}(1-\lambda)^{3-1}\lambda^1 &&= 3\lambda(1-\lambda)^2 \\
b_{3,2}(\lambda) &= \binom{3}{2}(1-\lambda)^{3-2}\lambda^2 &&= 3\lambda^2(1-\lambda) \\
b_{3,3}(\lambda) &= \binom{3}{3}(1-\lambda)^{3-3}\lambda^3 &&= \lambda^3
\end{aligned}$$

$b_{3,i}(\lambda)$ with $i = 0, 1, 2, 3$ are the cubic Bernstein polynomials of degree 3.

Definition. Given are the points

$$Q_i = \begin{pmatrix} x_i \\ y_i \end{pmatrix} \quad \text{mit} \quad Q_i \in \mathbb{R}^2, \quad i = 0, 1, \ldots, n$$

A **Bézier curve** is then defined by

$$C(\lambda) = \sum_{i=0}^{n} Q_i \cdot b_{i,n}(\lambda) \tag{25.3}$$

The points $Q_i, i = 0, \ldots, n$ are called **control points**.

The control points of a Bézier curve form the so-called control polygon.

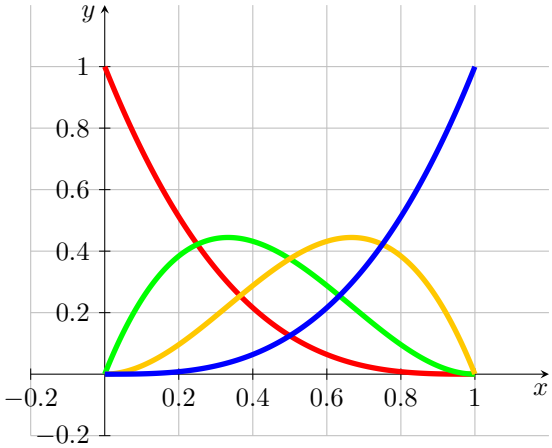


Figure 25.1.: Bernstein polynomial of degree 3

Bemerkung. Let the starting point P_0 and the end point P_1, as well as the tangents $\vec{t}_0$ and $\vec{t}_1$ be given. The tangents are not necessarily normalised.
The control points Q_0, Q_1, Q_2 and Q_3 of the associated Bézier curve are given by the following equations:

$$Q_0 = P_0, \qquad Q_1 = P_0 + \lambda_0 \vec{t}_0, \qquad Q_2 = P_1 - \lambda_1 \vec{t}_n, \qquad Q_3 = P_1 \tag{25.4}$$

[Jak+10]

Beispiel. Given are

$$P_0 = \begin{pmatrix} 2 \\ 4 \end{pmatrix}, \quad \vec{t}_0 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \text{und} \quad P_1 = \begin{pmatrix} 6 \\ 1 \end{pmatrix}, \quad \vec{t}_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

Then, according to theorem **??** for control points of the associated Bézier curve results:

$$Q_0 = P_0 = \begin{pmatrix} 2 \\ 4 \end{pmatrix}, \qquad Q_1 = P_0 + \vec{t}_0 = \begin{pmatrix} 2 \\ 4 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 5 \end{pmatrix},$$

$$Q_2 = P_1 - \vec{t}_n = \begin{pmatrix} 6 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 5 \\ 3 \end{pmatrix}, \qquad Q_3 = P_1 = \begin{pmatrix} 6 \\ 1 \end{pmatrix}$$

The Bézier curve and its control points are shown in the figure **??**.

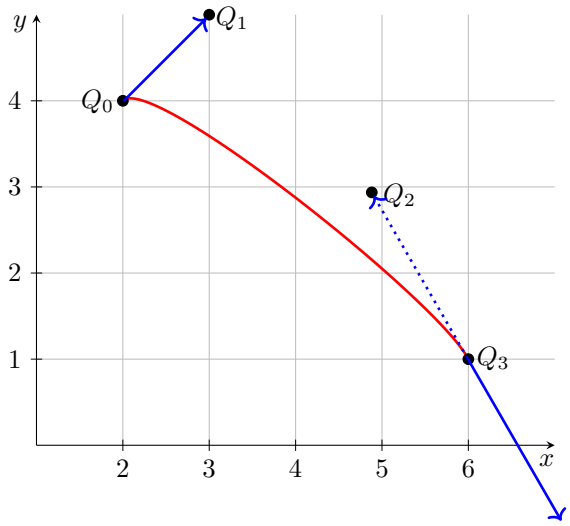


Figure 25.2.: Bézier curve for example 25.1

26. Verrundung mit einem Kreisbogen

26.1. Gleichungen

Blending with an arc is a simple variant of smoothing corners. This variant enables the processing and the generation of files according to DIN 66025. For smoothing, three points P_0 and S and P_1 are always considered, thus a symmetric Hermite problem results. According to theorem **??** it is assumed to be in the standard form $HP(L, \alpha)$.

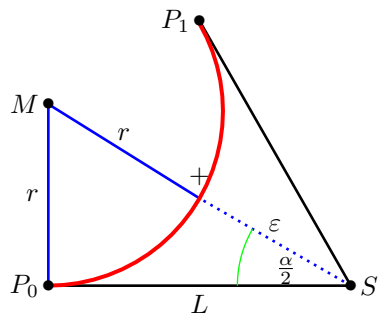


Figure 26.1.: Smoothing a corner with the help of an arc - triangle

The figure **??** represents the situation. The points P_0, S and P_1 are given. The included angle $\alpha = \angle(P_0; S; P_1)$ as well as the distance $L = \|\|S - P_0\|\| = \|\|P_1 - S\|\|$ are shown in the graph. The red arc is the desired result. The distance of its centre M to S is then $r + \varepsilon$, where r is the radius of the arc and ε is the given tolerance. Thus we obtain a right triangle P_0SM whose edge lengths are L, $r + \varepsilon$ and r.

According to the definition of the sine and the tangent, it follows:

$$\sin\left(\frac{\alpha}{2}\right) = \frac{L}{r+\varepsilon} \qquad \text{und} \qquad \tan\left(\frac{\alpha}{2}\right) = \frac{L}{r}$$

$$\Leftrightarrow \quad \varepsilon = \frac{L}{\sin\left(\frac{\alpha}{2}\right)} - r \qquad \text{und} \qquad r = \cot\left(\frac{\alpha}{2}\right) L$$

The two equations can be combined to give the following conditions:

$$\varepsilon = L \cdot \frac{1 - \cos\left(\frac{\alpha}{2}\right)}{\sin\left(\frac{\alpha}{2}\right)} \qquad \text{bzw.} \qquad L = \varepsilon \cdot \frac{\sin\left(\frac{\alpha}{2}\right)}{1 - \cos\left(\frac{\alpha}{2}\right)}$$

This gives the equation for the radius:

$$r = L \cdot \cot\left(\frac{\alpha}{2}\right) = L \cdot \sqrt{\frac{1 - \cos(\alpha)}{1 + \cos(\alpha}} = L \cdot \frac{\sin(\alpha)}{1 + \cos(\alpha)} = L \cdot \frac{P_{1,x}}{P_{1,y}}$$

The factor for converting L and ε can be simplified using trigonometric transformations.

$$\frac{1-\cos\left(\frac{\alpha}{2}\right)}{\sin\left(\frac{\alpha}{2}\right)} = \frac{1-\sqrt{\frac{1-\cos(\alpha)}{2}}}{\sqrt{\frac{1+\cos(\alpha)}{2}}} = \frac{\sqrt{2}-\sqrt{1-\cos(\alpha)}}{\sqrt{1+\cos(\alpha)}} = \frac{\sqrt{1+\cos(\alpha)}-\sin(\alpha)}{1+\cos(\alpha)}$$

By comparing this with the symmetric Hermite problem in standard form, the following notation is then obtained:

$$\frac{1-\cos\left(\frac{\alpha}{2}\right)}{\sin\left(\frac{\alpha}{2}\right)} = \frac{\sqrt{P_{1,x}}-P_{1,y}}{P_{1,x}}$$

The previous considerations are now summarised in the following sentence.

Satz. Let a symmetric Hermite problem $(P_0,\, \vec{t}_0,\, P_1,\, \vec{t}_1, S, L)$ be given. Without restriction of generality, it is in the standard form

$$\left\{\begin{pmatrix}0\\0\end{pmatrix}, \begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}L+L\cdot\cos(\alpha)\\L\cdot\sin(\alpha)\end{pmatrix}, \begin{pmatrix}\cos(\alpha)\\\sin(\alpha)\end{pmatrix}, \begin{pmatrix}L\\0\end{pmatrix}, L\right\}$$

with $L \in \mathbb{R}^{>0}$ uad $\alpha \in (-\pi;\, \pi]$.

Then an arc can be found that connects the points P_0 and P_1, has the same tangent directions at the two points.

For the circular arcs applies:

$$r = L\cdot\left|\frac{P_{1,x}}{P_{1,y}}\right|; \quad \phi_0 = -\operatorname{sign}(\alpha)\cdot\frac{\pi}{2}; \quad \phi_1-\phi_0 = \operatorname{sign}(\alpha)\cdot\pi-\alpha = \beta;$$

$$M = P_0 + \operatorname{sign}(\alpha)\cdot r\cdot\vec{t}_0^{\perp} = \operatorname{sign}(\alpha)\cdot r\cdot\begin{pmatrix}0\\1\end{pmatrix}$$

From the specification of the distance L, the maximum error can be calculated, as shown in theorem **??**. According to the derivation, it is also possible to specify the maximum error ε and determine the maximum distance L from it.

Satz. Let a symmetric Hermite problem $(P_0,\, \vec{t}_0,\, P_1,\, \vec{t}_1, S, L)$ be given. Without restriction of generality, it is in the standard form

$$\left\{\begin{pmatrix}0\\0\end{pmatrix}, \begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}L+L\cdot\cos(\alpha)\\L\cdot\sin(\alpha)\end{pmatrix}, \begin{pmatrix}\cos(\alpha)\\\sin(\alpha)\end{pmatrix}, \begin{pmatrix}L\\0\end{pmatrix}, L\right\}$$

with $L \in \mathbb{R}^{>0}$ and $\alpha \in (-\pi;\, \pi]$.

a) Let the maximum error ε be given. The maximum distance L for which an arc exists according to the theorem **??** that takes the error into account is given by:

$$L(\varepsilon,\alpha) = \varepsilon\cdot\frac{P_{1,x}}{\sqrt{P_{1,x}}-P_{1,y}} = \varepsilon\cdot\frac{L+L\cdot\cos(\alpha)}{\sqrt{L+L\cdot\cos(\alpha)}-L+L\cdot\cos(\alpha)}$$

b) When the distance L is specified, the following maximum error results:

$$\varepsilon(L,\alpha) = L\cdot\frac{\sqrt{P_{1,x}}-P_{1,y}}{P_{1,x}} = L\cdot\frac{\sqrt{L+L\cdot\cos(\alpha)}-L+L\cdot\cos(\alpha)}{L+L\cdot\cos(\alpha)}$$

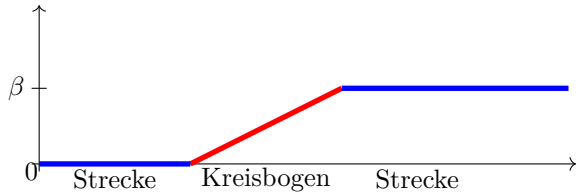


Figure 26.2.: Angular change with blending arc

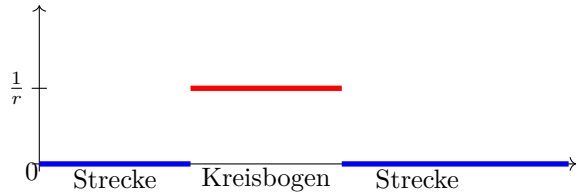


Figure 26.3.: Curvature progression for a blending arc

These considerations result in limitations for the use of this strategy, which are summarised in the following comment.

Bemerkung. The following conditions for applying the strategy of a circle must be fulfilled:

a)
$$P_0 \neq P_1$$

b)
$$\vec{t_0} = -\vec{t_1} \quad \Leftrightarrow \quad \alpha = 0$$

c)
$$\vec{t_0} = \vec{t_1} \quad \Leftrightarrow \quad \alpha = \pi$$

26.2. Bewertung

Rounding by means of a circular arc leads to a continuous course of the angle change. The figure **??** illustrates this.

Due to the rounding with a circular arc, the curvature of the curve is not continuous. The figure **??** shows that the curvature is constant in the range of distances 0 and on the circular arc with the value $\frac{1}{r}$, where r is the radius of the circular arc used. Due to the formula $a = \frac{v^2}{r}$ for a movement on a circular arc, the acceleration course exhibits continuity jumps at the transitions for a constant path velocity.

Depending on the angle change β at the corner S and the tolerance ε, the maximum length of the shortening of the lines can be calculated. The figure **??** shows the corresponding diagram, where the tolerance ε is set to the value 1.

The area provided can also be specified. Depending on the distance L from the corner point S, the deviation ε can be calculated. The figure **??** represents the function as a function of L and the angle α. The figures **??** and **??** show the curves of the length as a function of the angle change for a fixed tolerance and the error as a function of the angle change for a given length. If the angle change is minimal, then

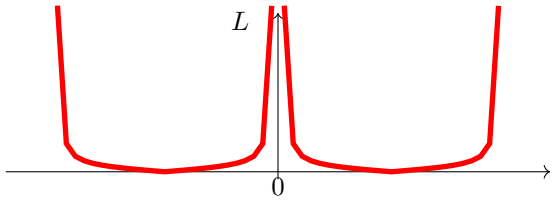


Figure 26.4.: Maximum range for a blending arc of a circle - $L(\varepsilon = const, \alpha)$

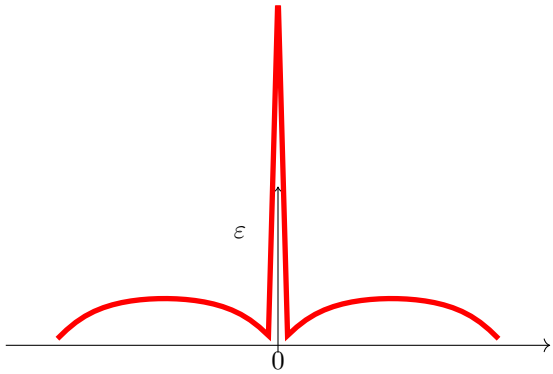


Figure 26.5.: Deviation when specifying the distance L when rounding with a circular arc

the error or length L is extreme. In this case, rounding by means of an arc is not possible. In order to develop a sensible strategy, a minimum angle change must be specified from which a blemnding arc is permitted. Likewise, a maximum angle change must also be defined. For an angular change of $\pm\pi$ is a bend. In this case, a blending is also not possible.

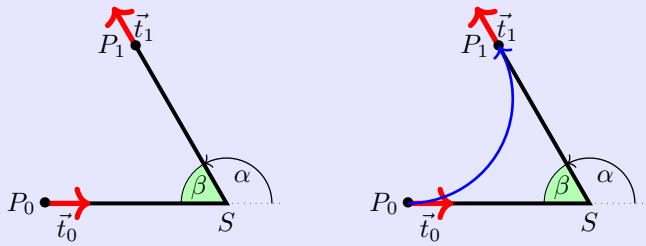
26.3. Examples

Beispiel. Let it be the symmetric Hermite problem

$$\left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 6.0 + 6.0 \cdot \cos\left(\frac{2}{3}\pi\right) \\ 6.0 \cdot \sin\left(\frac{2}{3}\pi\right) \end{pmatrix}, \begin{pmatrix} \cos\left(\frac{2}{3}\pi\right) \\ \sin\left(\frac{2}{3}\pi\right) \end{pmatrix}, \begin{pmatrix} 6.0 \\ 0 \end{pmatrix}, 6.0 \right\}$$

with $L = 6$ and $\alpha = \frac{2}{3}\pi$.
For the blending arc follows:

$$M = \begin{pmatrix} 0 \\ 3,4641 \end{pmatrix}; \quad r = 3,46412; \quad \phi_0 = -\frac{\pi}{2}; \quad \alpha = \frac{2}{3}\pi$$

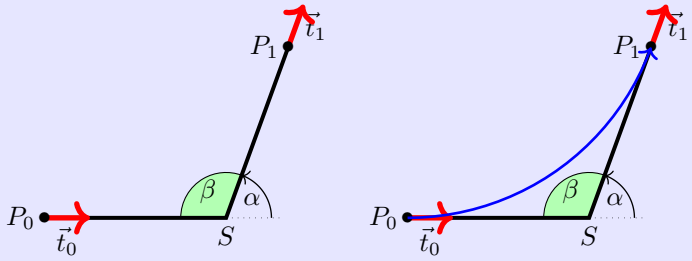


Beispiel. Let it be the symmetric Hermite problem

$$\left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 6.0 + 6.0 \cdot \cos\left(\frac{7}{18}\pi\right) \\ 6.0 \cdot \sin\left(\frac{7}{18}\pi\right) \end{pmatrix}, \begin{pmatrix} \cos\left(\frac{7}{18}\pi\right) \\ \sin\left(\frac{7}{18}\pi\right) \end{pmatrix}, \begin{pmatrix} 6.0 \\ 0 \end{pmatrix}, 6.0 \right\}$$

with $L = 6$ and $\alpha = \frac{7}{18}\pi$.
For the blending arc follows:

$$M = \begin{pmatrix} 0 \\ 8,5689 \end{pmatrix}; \quad r = 8,5689; \quad \phi_0 = -\frac{\pi}{2}; \quad \alpha = \frac{7}{18}\pi$$

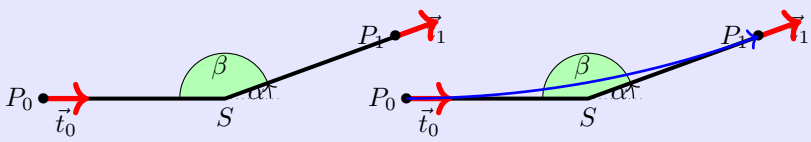


Beispiel. Let it be the symmetric Hermite problem

$$\left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 6.0 + 6.0 \cdot \cos\left(\frac{1}{9}\pi\right) \\ 6.0 \cdot \sin\left(\frac{1}{9}\pi\right) \end{pmatrix}, \begin{pmatrix} \cos\left(\frac{1}{9}\pi\right) \\ \sin\left(\frac{1}{9}\pi\right) \end{pmatrix}, \begin{pmatrix} 6.0 \\ 0 \end{pmatrix}, 6.0 \right\}$$

with $L = 6$ and $\alpha = \frac{1}{9}\pi$.
For the blending arc follows:

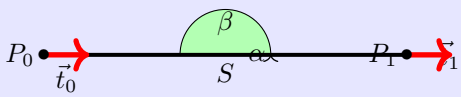
$$M = \begin{pmatrix} 0 \\ 34,0277 \end{pmatrix}; \quad r = 34,0277; \quad \phi_0 = -\frac{\pi}{2}; \quad \alpha = \frac{1}{9}\pi$$



Beispiel. Let it be the symmetric Hermite problem

$$\left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 12.0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 6.0 \\ 0 \end{pmatrix}, 6.0 \right\}$$

with $L = 6$ and $\alpha = 0$.
There is no blending arc here.

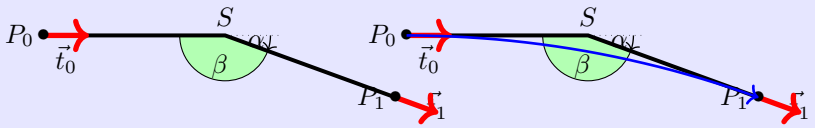


Beispiel. Let it be the symmetric Hermite problem

$$\left\{\begin{pmatrix}0\\0\end{pmatrix}, \begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}6.0+6.0\cdot\cos\left(-\frac{1}{9}\pi\right)\\6.0\cdot\sin\left(-\frac{1}{9}\pi\right)\end{pmatrix}, \begin{pmatrix}\cos\left(-\frac{1}{9}\pi\right)\\\sin\left(-\frac{1}{9}\pi\right)\end{pmatrix}, \begin{pmatrix}6.0\\0\end{pmatrix}, 6.0\right\}$$

with $L = 6$ and $\alpha = -\frac{1}{9}\pi$.
For the blending arc follows:

$$M = \begin{pmatrix}0\\-34,0277\end{pmatrix}; \quad r = 34,0277; \quad \phi_0 = \frac{\pi}{2}; \quad \alpha = \frac{1}{9}\pi$$

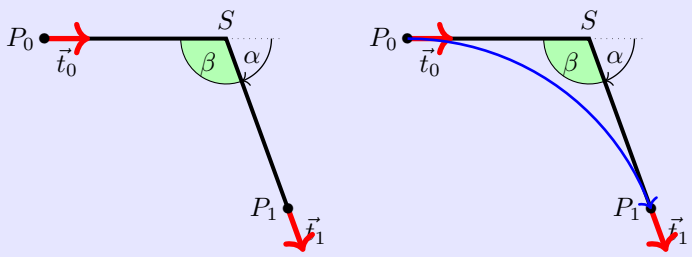


Beispiel. Let it be the symmetric Hermite problem

$$\left\{\begin{pmatrix}0\\0\end{pmatrix}, \begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}6.0+6.0\cdot\cos\left(-\frac{7}{18}\pi\right)\\6.0\cdot\sin\left(-\frac{7}{18}\pi\right)\end{pmatrix}, \begin{pmatrix}\cos\left(-\frac{7}{18}\pi\right)\\\sin\left(-\frac{7}{18}\pi\right)\end{pmatrix}, \begin{pmatrix}6.0\\0\end{pmatrix}, 6.0\right\}$$

with $L = 6$ and $\alpha = -\frac{7}{18}\pi$.
For the blending arc follows:

$$M = \begin{pmatrix}0\\-8,5689\end{pmatrix}; \quad r = 8,5689; \quad \phi_0 = \frac{\pi}{2}; \quad \alpha = -\frac{7}{18}\pi$$

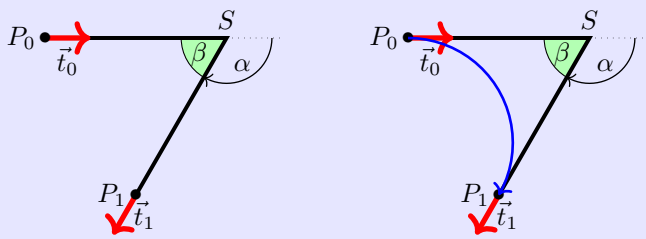


Beispiel. Let it be the symmetric Hermite problem

$$\left\{\begin{pmatrix}0\\0\end{pmatrix}, \begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}6.0+6.0\cdot\cos\left(-\frac{2}{3}\pi\right)\\6.0\cdot\sin\left(-\frac{2}{3}\pi\right)\end{pmatrix}, \begin{pmatrix}\cos\left(-\frac{2}{3}\pi\right)\\\sin\left(-\frac{2}{3}\pi\right)\end{pmatrix}, \begin{pmatrix}6.0\\0\end{pmatrix}, 6.0\right\}$$

with $L = 6$ and $\alpha = -\frac{2}{3}\pi$.
For the blending arc follows:

$$M = \begin{pmatrix}0\\-3,4641\end{pmatrix}; \quad r = 3,46412; \quad \phi_0 = \frac{\pi}{2}; \quad \alpha = -\frac{2}{3}\pi$$



27. Maple files

[Wat17c; Wat17d; Wat17b; Wat17e; Wat17a; Wat17f]

27.1. Geometries with Maple

The algorithms presented can be well represented using geometries. Two aspects can be investigated. On the one hand, the effort for an implementation, the computational accuracy and the stability of an algorithm are interesting; on the other hand, the methods are to be evaluated and compared with regard to their usefulness. For this purpose, modules for the formula manipulation system Maple [Wat17c] were developed.
Maple offers the environment to implement algorithms easily and quickly. Furthermore, graphical representation is possible with simple means. However, a simple workbook of Maple is not designed for very large software projects. Here, one must resort to the possibility of creating and using libraries. On the one hand, Maple offers the possibility of creating one's own libraries, so-called modules. In this way, one remains within the environment and syntax of Maple. This path will be pursued further here. Another possibility is the use of libraries created by means of a higher programming language, e.g. C++, the so-called DLLs.
In the following, the creation and use of a test environment with the help of modules is described. First, the geometry elements that are stored in various modules and their use are described. The structure and use of a module in Maple is then explained so that own extensions and additions are possible.

27.2. Geometry elements used

This is a test environment for the algorithms presented, only geometries that have been described are also used.

- points (`MPoint`)
- lines (`MLine`)
- arcs (`MArc`)
- Bézier curves (`Bezier`)
- polygon courses (`MPolygon`)
- Geometry List (`MGeoList`)
- Hermite problems (`MHermiteProblem`)
- Symmetric Hermite Problems (`MHermiteProblemSym`)

For each geometry element a corresponding module has been created, the name of which is given in the list above. The list of which functions are available is presented in another section.
When implementing such a project, it quickly becomes clear that when using several geometry elements, a clear data structure and well-defined access to the data is essential. For example, when representing straight lines, the decision must be made whether the representation should be point-directional or by means of two points. The choice is generally made depending on the application. Here, the path is taken that, due to a well-defined access to the data, the use of both representations is possible.

27.3. Building the data structure

The idea is to use a data structure that is as uniform and simple as possible. Maple does offer the possibility to define objects. However, it is difficult to use within a procedural environment. Therefore, all data is basically represented as lists. The first list element always contains an identifier of the element **??**. This is followed by the data, which in turn can be geometry elements. It should be noted that direct access to the data cannot be prevented; here the user is responsible.

Access to the data should be exclusively via procedures of a module. The following is an example of the data structure for points:

```
[MVPOINT,[x,y]]
```

The creation of a point then takes place via a procedure `New`:

```
NeuerPunkt := MPoint:-New(10,15);
```

When called up in the test file, the following is then output:

```
TestP0 := ["Point",[10,15]]
```

These data structures are used for the calculation of geometries. They are used in functions or procedures. Unlike variables, the data structures and procedures are defined globally and not locally. Under the command `export` the procedures are declared at the beginning of the module and can thus also be used outside the module. If, for example, a data structure from `MPoint` is to be used in another module or outside the modules, it is called as follows:

```
P0 := MPoint:-New(x,y);
```

In addition to the modules for the geometries, there is a module (MConstant) for saving constants and names. There, for `MVPOINT`, for example. „Point “ or e.g. a constant for the comparison to zero for real numbers.
Finally there is the test file, which is not a module, in which the modules are loaded, called and tested in their function. More about this file later in „1.4 Maple test file “.

27.4. Structure of a module

The following section deals with the purpose of modules and their structure.

The project is about capturing the above geometries in a data structure and representing them in Maple. However, the calculations and formulas that have to be used are a hindrance to the final representation. Modules are very well suited for summarising and hiding the calculations that take place in functions. This way, the important functions can be accessed in Maple without seeing what is in them.

Example:
The function `Angle` from the module `MPoint`: Function to calculate an angle between location vector and x-axis:

```
Angle := proc(P)
   local alpha, x, y;
   x := GetX(P);
   y := GetY(P);
   alpha := 0;
   if abs(x) < 0.00001
      then
         alpha := 3*Pi/2;
```

```
      else
         alpha := Pi/2;
      end if;
   else
      alpha := arctan(y/x);
      if x < 0
      then
         alpha := alpha + Pi;
      else
         if y < 0
         then
            alpha := alpha + 2*Pi;
         end if;
      end if;
   end if;
   return factor(alpha);
end proc;
```

This function is long, but it only represents a small part of the programme for the representation in question. That is why it is in the module and can thus be called externally with a single command:

```
Winkel := MPoint:-Angle(P)
```

The following section describes the structure of a module. Since Maple offers a wide range of possibilities, a restriction is made. Only the structure of the modules used is described, here on the basis of the moduleMPoint. For more detailed possibilities, please refer to the Maple manual [Wat17c].

<u>structure:</u>

The module must first be started. This is done with the name (here always a capital M and the geometry) of the module and the following command:

```
MPoint := module()
```

Then, similar to procedures, the variables and functions must be defined. You can declare them either locally or globally. If the variables or procedures are only used and changed in the module, the declaration is made with the command `local` as follows:

```
local Variable names, with, comma, separated;
```

If the variables or procedures are also to be usable outside the module, this is defined with `export`:

```
export Function names, with, comma, separated;
```

This is followed by the specification of the options:
`option package;`,
the description of the module:
`description “Self-selected module description, e.g. module for points “`;
and the initialisation of the module:

```
ModuleLoad := proc
  MVPOINT:=MConstant:-GetPoint();
  print("Module MPoint is loaded");
end proc;
```

From here on, programming is done as usual in Maple. The previously defined procedure and variable names are used and programming is done with commands that are also used outside of a module in Maple. It is important to know that with modules, each function can be called constantly, so the order of the functions is irrelevant.

To finally end the module, use the following command:

```
end module;
```

27.4.1. General structure of a module

In summary, the modules have the following structure:

```
Modulname := module()

   local Names, with, comma, separated;

   export Names, with, comma, separated;

   global Names, with, comma, separated;[1]

   option package;

   description „ Self-selected module description“;


   ModuleLoad := proc()[2]
      MVPOINT:=MConstant:-GetPoint();
      print("Modul MPoint is loaded");
   end proc;

 Procedure1 := proc (Passing parameters)
      inhalt;
   end proc;

   Procedure2 := proc (Passing parameters)
      content;
   end proc;


   ...

end module;
```

[1]Possible, but not used in the modules
[2]Example `MPoint`

27.4.2. Saving a module

The management of modules is done automatically by Maple. However, since the modules are passed on and have to be edited individually, some settings have to be made. Therefore, the following prefix is used for each Maple file of the project:

```
restart;
with(LibraryTools);
lib := "C:/FH/Tools/Maple/MyLibs/Blending.mla";
march('open', lib);
ThisModule := 'MArc';
```

The first line initialises the system. The next 3 lines enable working with archives. In the second line, the tools are loaded so that the variable `lib` can be occupied. All modules are stored in an archive; in this example it is the file `Blending.mla` in the directory `C:/FH/Tools/Maple/MyLibs/`. The command `ThisModule := 'MArc';` contains the name of the current module.

A module is then saved using the command `savelib('ModuleName')`. A file `ModuleName.mla` is then created. Depending on the configuration of Maple, the set path where the file is automatically created may not be writable. Then you can configure the system so that the mla file is stored in the current directory or in a directory of your choice.

If the above prefix is used for a Maple file, all that is required to save the module is
`savelib(ThisModule, lib);`

27.4.3. Verwendung eines Moduls

A module that has been saved can now be used in other Maple worksheets. To do this, the command
`with(ModuleName)`
is used. If you have used a special path, Maple cannot find the file `ModuleName.mla`. Then the path must be made known, e.g.:
`savelibname := "c:/Maple/MyLibs";", savelibname;`
Now the procedures of the module can be accessed. An overview of all exported procedures is provided by the command
`Describe(ModuleName)`
is displayed. A list of the names including the names of the transfer parameters is displayed. If a procedure is equipped with the field `description`, this text is also displayed.

27.4.4. Creating a Maple module

Creating your own module is done quickly if you use the framework above. However, one should make some considerations beforehand and maintain a standard.

Before creating an own module, the data model and the corresponding procedures should be worked out. A programme flow chart is useful here. The central task of the module is usually quickly determined. In addition, however, the following rules should be observed.

Rule 1 Basically, both the module and each procedure receive a description. This is not limited to the optional argument `description`, but is placed in front of each procedure. The description basically contains the description of the task. The prerequisite is also mentioned or an error handling is described. Then follows the description of all input parameters, their function and their data structure. The return value is then described.

rule 2 Each module receives a procedure `Version()`, which returns the current version number.

rule 3 Data is not global. To access data, procedures `Set*` and `Get*` are provided. These procedures are also used within the procedures of a module.

rule 4 At least one test function is written for each procedure. The test function can be used to illustrate the use and any special features.

27.5. Functions of the modules

In the following, the individual modules are listed. The data structure of each module and all functions with associated tasks are listed.

27.5.1. Module MPoint

MPoint is a module for points and works with a data structure and with procedures/functions. The data structure New for a point is a list, which is structured as follows:

```
[MVPOINT, [x,y]]
```

Its first element is a name to identify the module. Your second element is again a list containing the elements of the geometry. In this case the name is "Point" and the elements are the x and y coordinates for a point. No distinction is made here between point and vector. A point can also be seen as a vector from the coordinate origin to the point.
Via the Get functions GetX and GetY one can capture the coordinates of the point and use them in other functions. The other functions can then be used to calculate with the points/vectors.

MPoint		
E	**New**	Data structure for a point
E	**GetX**	Reading the x-coordinate
E	**GetY**	reading the y-coordinate
E	**Angle**	calculating the angle between the x-axis and the point
E	**Add**	Calculates a linear combination of two points/vectors
E	**Sub**	Calculates the difference of two points/vectors
E	**Cos**	Calculates the cosine between two vectors
E	**Sin**	Calculates the sine between two vectors
E	**Scale**	Scales a vector with a factor
E	**Perp**	Calculates a vector that is orthogonal to the given vector
L	**IllustrateXY**	Plot function to illustrate a blue point
E	**Illustrate**	Plot of a blue point
E	**Plot**	Plot of a green point
E	**Plot2D**	Plot of a point with own options
E	**Length**	Calculates the distance of the point to the coordinate origin
E	**Uniform**	Normalises the vector to length 1
E	**LinetoVector**	Calculates vector from a distance
E	**Distance**	Calculates the distance between two points

27.5.2. Module MLine

MLine is a module for lines and works with a data structure and with procedures/functions. The data structure New for a line is a list which is structured as follows:

```
[MVLINE, [P0,P1]]
```

Its first element is a name to identify the module. Its second element is again a list containing the elements of the geometry. In this case the name is „Line“ and the elements are the start and end points for a line.
The data structure NewPointVerctor is also a data structure for a line, but here the line is defined by a start point and a direction vector.

The Get functions StartPoint and EndPoint can be used to capture the start and end points of the route and use them in other functions. The other functions can then be used to calculate with the routes.

MLine		
E	**New**	Data structure for a line (two-point form)
E	**NewPointVector**	creation of a route (point-direction form)
E	**StartPoint**	reading the starting point
E	**EndPoint**	reading the end point
E	**Position**	Calculate a point that lies on the line
E	**Plot2D**	Plot a part of the route starting from the starting point
E	**Plot2DTangent**	Plot of a point with tangent
E	**Plot2DTangentArrow**	Plot of a point with tangent (as arrow)
E	**LineLine**	Calculation of the intersection of two straight lines.
E	**AngleLine**	Calculation of the angle between two straight lines

27.5.3. Module MArc

MArc is a module for circular arcs and works with a data structure and with procedures/functions. The data structure New for an arc is a list that is structured as follows:

```
[MVARC, [mx,my,r,phi,alpha]]
```

Its first element is a name to identify the module. Your second element is again a list containing the elements of the geometry. In this case the name is "Arc" and the elements are the x- and y-coordinate for the centre, the radius, the start angle and the angle change for an arc.

The Get functions GetM, GetMX, GetMY, GetR, GetPhi, and GetAlpha can be used to collect the data for an arc and use it in other functions. The other functions can then be used to calculate with the arcs.

MArc		
E	**New**	Data structure for an arc
E	**GetMX**	Reading the x-coordinate of the centre point.
E	**GetMY**	read the y-coordinate of the centre point
E	**GetR**	read the radius
E	**GetPhi**	reading the start angle
E	**GetAlpha**	Reading the change of angle
E	**GetM**	reading the centre point
E	**Position**	Calculating a point on the arc
E	**Plot2D**	Plot a part of the arc starting from the start angle
E	**Blend**	calculating an arc from a symmetric Hermite problem

27.5.4. module MBezier

The New data structure for a polygon is a list constructed as follows:

```
[MVBEZIER, [PointList]]
```

Within the module MBezier exist the procedures listed in the following table.

MBezier		
E	New	Manual input of control points for a Bézier curve
E	Version	Output of the verions
E	BlendCurvature	Determination of control points from symmetric Hermite problem
E	BlendCurvatureEpsilon	determination of control points from symmetric Hermite problem with given error
E	Position	position on the Bézier curve
E	GetTheta	Reading the angle
E	GetEpsilon	reading the tolerance
E	GetControlPoint	Read the control points
E	Plot2D	Read the Bézier curve
E	PlotControlPoints	representation of all control points

27.5.5. Module MPolygon

MPolygon is a module for polygons and works with a data structure and with procedures/functions. The data structure New for a polygon is a list that is structured as follows:

```
[MVPOLYGON, [PointList]]
```

Its first element is a name to identify the module. Your second element is again a list containing the elements of the geometry. In this case the name is "Polygon" and the elements are any number of points. Using the Get functions GetPoint and GetN you can get the number of points and the points themselves and use them in other functions. The other functions can then be used to calculate with the points or the polygon course.

MPolygon		
E	New	Data structure for a list of points
E	GetPoint	Reading the ith point from the point list
E	GetN	Determine the number of points in the point list
E	Length	Determination of the Euclidean length of the polygon
E	Position	Calculation of a point on the polygon course
E	Tangents	calculation of the tangent
L	Plot2DAll	Plot list of all points
E	Plot2D	Plot of all points as polygonal plots.
E	Plot2DTangent	representation of the polygon course with tangent
E	PlotPoints	representation of all points

27.5.6. module MGeoList

MeoList is a module for a geometry list and works with a data structure and with procedures/functions. The data structure New for a geometry list is a list that is structured as follows:

```
[MVGEOLIST, []]
```

Its first element is a name to identify the module. Its second element is again a list containing the elements of the geometry. In this case the name is „GeoList“ and the elements are any number of individual geometries.
Using the Get functions GeoGeo and GetN, the i-th element of the list and the number of elements in the list can be captured and used in other functions. The other functions can then be used to calculate with the geometries or the list. In the functions Length, Plot2DAll, Plot2D and Position the functions are called in themselves. This is possible because the functions work independently of each other.

MGeoList		
E	New	Data structure for a geometry list
E	Append	Append a geometry element to the data structure
E	Prepend	Inserting a geometry element as the first element of the list
E	Replace	Replace a geometry element with another one.
E	GetN	Determine the number of geometry elements in the list
E	GeoGeo	Reading the i-th geometry element
E	Length	Calculate the Euclidean length of the geometry elements
E	Position	Calculates a point on the geometry
L	Plot2DAll	Plot function for plotting the geometry elements
E	Plot2D	Plot a part of the geometry list starting from the first element

27.5.7. Module MHermiteProblem

MHermiteProblem is a module for Hermite problems and works with a data structure and with procedures/functions. The data structure New for a route is a list, which is structured as follows:

```
[MVHERMITEPROBLEM, [P0,T0n,P1,T1n]]
```

Your first element is a name to identify the module. Its second element is again a list containing the elements of the geometry. In this case the name is „HermiteProb“ and the elements are two points with associated tangents (lines).

Using the Get functions StartPoint, EndPoint, StartTangent and EndTangent, the points and their tangents can be captured and used in other functions. The other functions can then be used to calculate with the data for the Hermite problem.

MHermiteProblem		
E	New	Data structure for a Hermite problem
E	StartPoint	reading the start point
E	EndPoint	Reading the end point command
E	StartTangent	reading the start tangent
E	EndTangent	Reading the end tangent
E	Plot2D	Plotting the Hermite problem

27.5.8. Module MHermiteProblemSym

MHermiteProblemSym is a module for symmetric Hermite problems and works with a data structure and with procedures/functions. The data structure New for a symmetric Hermite problem is a list built as follows:

```
[MVHERMITEPROBLEMSYM, [P0,T0n,P1,T1n,S,L]]
```

Your first element is a name to identify the module. Its second element is again a list containing the elements of the geometry. In this case the name is „SymHermiteProb“ and the elements are two points with associated tangents, their intersection, and the distance of the points to the intersection.

Via the Get functions StartPoint, EndPoint, StartTangent, EndTangent, CrossPoint and ParameterL one can acquire the data and use it in other functions. The other functions can then be used to calculate with the data for the symmetric Hermite problem.

MHermiteProblemSym		
E	New	Data structure for a symmetric Hermite problem
E	StartPoint	reading the start point
E	EndPoint	Reading the end point command
E	StartTangent	reading the start tangent
E	EndTangent	Reading the end tangent
E	ParameterL	reading of the distance
E	CrossPoint	reading of the intersection point
E	Plot2D	Plotting of the symmetrical Hermite problem
E	Create	creation of a Sym. Hermite problem by 3 points
E	BlendArc	rounding of the corner point by an arc

27.5.9. Module MConstant

MConstant is a module for storing fixed constants and names to identify data structures. In the locally declared functions, starting with CV, the constants for declaring the different geometries are defined. These are names for recognising the geometry elements in the test file. In the globally declared functions, starting with Get, the names for identifying the geometry elements are returned.

MConstant		
L	NULLEPS	constant to compare to zero
L	CVPOINT	constant for geometry elements: Point
L	CVLINE	constant for geometry elements: Line
L	CVARC	constant for geometry elements: Arc
L	CVPOLYGON	constant for geometry elements: polygon
L	CVGEOLIST	constant for geometry elements: GeoList
L	CVHERMITEPROBLEM	constant for geometry elements: HermiteProb
L	CVHERMITEPROBLEMSYMMETRIC	Const. for geometry elements: SymHermiteProb
L	CVBIARC	Const. for geometry elements: Biarc
E	GetNullEps	return of the zero comparison command.
E	GetPoint	identifier for points
E	GetLine	Identifier for lines
E	GetArc	identifier for circular arcs
E	GetPolygon	identifier for polygons
E	GetGeoList	Identifier for geometry lists
E	GetHermiteProblemSymmetric	Identifier for symmetric Hermite problems
E	GetHermiteProblem	ID for Hermite problems
E	GetBiarc	identifier for Biarcs

27.5.10. Module MGeneralMath

MGeneralMath is a module for general mathematical functions and works with the data structures and procedures.

MeneralMath		
E	MPoint	Data structure for a point
E	MPointX	Reading of the x-coordinate for a point
E	MPointY	reading the y-coordinate for one point
L	MPointIllustrateXY	plot structure for a blue point
E	MPointIllustrate	plot structure for a blue point
E	MPointPlot	Illustration of a green point
E	MLine	Data structure for a line
E	MLineStartPoint	Reading of the start point for a draw frame
E	MLineEndPoint	Reading the end point for a line
E	MPointOnLine	Calculation of a point on the line
E	MLinePlot2D	Plot the part of a line starting at the starting point
. E	MLineLine	calculating the intersection of two linesline

27.5.11. Module Biarc

Data Structure

Requirements and specifications:

The Biarc is to be used to solve a Hermite problem. A Hermite problem is defined as follows:

Given two points P_0 and P_1 with associated normalised tangents $\vec{t_0}$ and $\vec{t_1}$. The biarc must connect the points such that the tangents of the start and end points of the biarc coincide with the tangents of the Hermites problem.

Data structure:

The data structure New for a biarc must contain two arcs.

So the data structure for one arc is needed: MArc:-New.

This in turn contains the coordinates for the centre, the radius, the start angle and the angle change.

27.5.12. Module MBiarc

MBiarc is a module for Biarcs and works with a data structure and with procedures/functions. The data structure New for a Biarc is a list which is structured as follows:

```
[MVBIARC, [Arc0, Arc1]]
```

Its first element is a name to identify the module. Your second element is again a list containing the elements of the geometry. In this case the name is „Biarc“ and the elements are two arcs.

Using the Get functions GetArc0 and GetArc1 one can capture the two arcs for the biarc. The functions listed below can be used to calculate the data for the biarc.

MBiarc		
E	New	Data structure for a biarc
E	GetArc0	Reading of the first arc
E	GetArc1	Reading the second arc
E	Circle	calculating the circle K_J from the Hermite problem data.
E	Plot2DCircle	plot of the circle K_J
E	angle	Calculate the angle from the centre of the circle to points on the circle
E	Plot2D	Plot of the biarc
E	ConnectionPoint	Calculation of the connection point J (Equal Chord)
E	TangentTj	Calculation of the tangent to J
E	Tangent Biarc	rotation of Tj for the biarc.
E	BiarcCenter	Calculate the centres of the arcs of the biarc
E	Biarc	calculate the centre of the biarc.
E	Position	Determine a point on the biarc
E	Blend	Calculate the biarc only from the Hermite problem

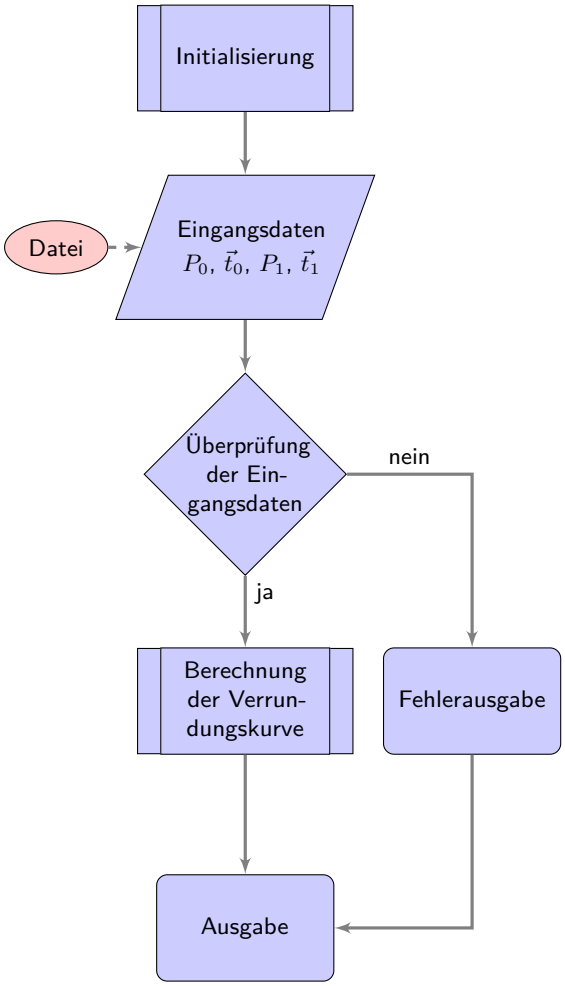


Figure 27.1.: Programme flow chart „Corner rounding“

27.6. Programme Flowchart

27.6.1. Overall Flow

27.6.2. Verrundung der Kurve

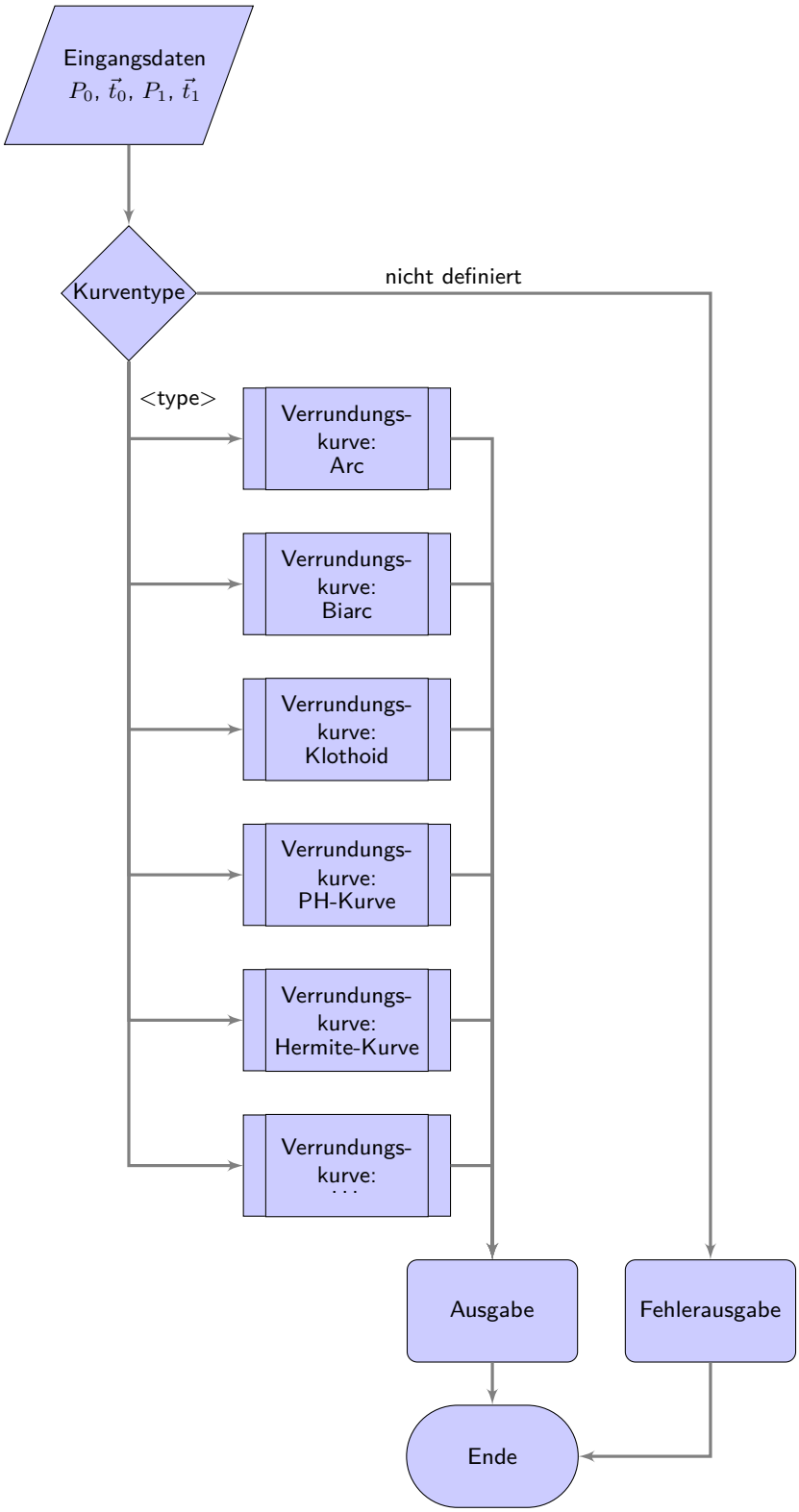


Figure 27.2.: Programme flowchart „Selection of the rounding strategies“

28. Representation of Python Programs

It is possible to integrate a part into the document, see 28.1. This is the most elegant method and is also preferable. Individual lines and line ranges can also be selected in the list of options. If a file is integrated, it is always as up-to-date as the document.
It may also be useful to integrate program lines: `redLED = pyb.LED(1)`.

Attention: The integration of programs with the help of images is pointless.

Listing 28.1.: The program “Hello World” in Python for microcontroller boards is inserted from the file `Blink.py`.

```
%%
%% This is file `rename-to-empty-base.tex',
%% generated with the docstrip utility.
%%
%% The original source files were:
%%
%% fileerr.dtx  (with options: `return')
%%
%% This is a generated file.
%%
%% The source is maintained by the LaTeX Project team and bug
%% reports for it can be opened at https://latex-project.org/bugs/
%% (but please observe conditions on bug reports sent to that address!)
%%
%%
%% Copyright (C) 1993-2024
%% The LaTeX Project and any individual authors listed elsewhere
%% in this file.
%%
%% This file was generated from file(s) of the Standard LaTeX `Tools Bundle'.
%% ---------------------------------------------------------------------------
%%
%% It may be distributed and/or modified under the
%% conditions of the LaTeX Project Public License, either version 1.3c
%% of this license or (at your option) any later version.
%% The latest version of this license is in
%%    https://www.latex-project.org/lppl.txt
%% and version 1.3c or later is part of all distributions of LaTeX
%% version 2005/12/01 or later.
%%
%% This file may only be distributed together with a copy of the LaTeX
%% `Tools Bundle'. You may however distribute the LaTeX `Tools Bundle'
%% without such generated files.
%%
%% The list of all files belonging to the LaTeX `Tools Bundle' is
%% given in the file `manifest.txt'.
%%
 \message{File ignored}
\endinput
%%
%% End of file `rename-to-empty-base.tex'.
```

../../Code/HelloWorld/Blink.py

29. Representation of Programs written for Arduino Boards

It is possible to integrate a part into the document, see 29.1. This is the most elegant method and is also preferable. Individual lines and line ranges can also be selected in the list of options. If a file is integrated, it is always as up-to-date as the document.

Attention: The integration of programs with the help of images is pointless.

Listing 29.1.: The program “Hello World” for an Arduino microcontroller boardis inserted from the file Blink.ino.

```
/**
*
*  @file Blink.ino
*
*  @brief Simple program for testing the configuration
*
*
*  Turns an LED on for one second, then off for one second, repeatedly.
*
*  Most Arduinos have an on-board LED you can control. On the UNO, MEGA and ZERO
*  it is attached to digital pin 13, on MKR1000 on pin 6. LED_BUILTIN is set to
*  the correct LED pin independent of which board is used.
*  If you want to know what pin the on-board LED is connected to on your Arduino
*  model, check the Technical Specs of your board at:
*
*  https://www.arduino.cc/en/Main/Products
*
*  modified 8 May 2014
*  by Scott Fitzgerald
*  modified 2 Sep 2016
*  by Arturo Guadalupi
*  modified 8 Sep 2016
*  by Colby Newman
*
*  This example code is in the public domain.
*
*  https://www.arduino.cc/en/Tutorial/BuiltInExamples/Blink
*/



/**
*  @brief the setup function runs once when you press reset or power the board
*
*  standard function of Arduino sketches
*
*  Initialization of the pin LED_BUILTIN as output
*
*  @param ---
*
*  @return void
*/
void setup() {
  // initialize digital pin LED_BUILTIN as an output.
  pinMode(LED_BUILTIN, OUTPUT);
}

/**
*  @brief the loop function runs over and over again forever
*
*  standard function of Arduino sketches
*
*  swichting the led on / off for 1sec.
*
*  @param ---
*
*  @return void
*/

void loop() {
  digitalWrite(LED_BUILTIN, HIGH);   // turn the LED on (HIGH is the voltage level)
  delay(1000);                       // wait for a second
  digitalWrite(LED_BUILTIN, LOW);    // turn the LED off by making the voltage LOW
  delay(1000);                       // wait for a second
}
```

../../Code/Arduino/Blink/Blink.ino

30. First Chapter

...
A Speicherprogrammierbare Steuerung (SPS) is ...

A Computerized Numerical Control (CNC) needs a SPS (PLC) to ...

31. CAGD

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.
The book CAGD by Gerald Farin is a classic on splines. [Far02]

The standard 66025 for programming CNC machines is also a classic; however, it does not deal with splines. [DIN66025-2]

Mr. F. Farouki has dealt with both CNC machine programming and splines. His article[1] on a real-time interpolator also shows this. [FS17]

A new dimension of machine tools have emerged with the invention of 3D printers. [Rus+07]

Another aspect of automation technology is communication. Another milestone has been reached with 5G technology. another milestone has been reached.[2]

[1] co-author is J. Srinathu
[2] Zaf+20.

A. drawings with tikz

The package tikz is a powerful tool for creating graphics. Many introductions exist. Here only the first steps are shown, so that you can easily create flowcharts.

Drawing a line and arrows

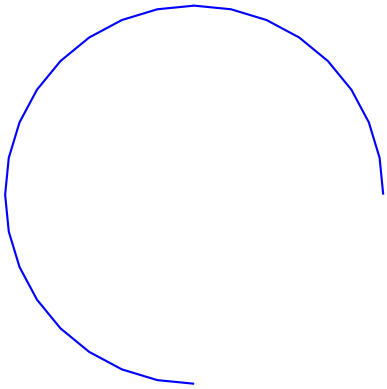
```
\begin{tikzpicture}
    \draw (0,0) -- (1,0);

    \draw[->] (2,0) -- (3,0);

    \draw[<->] (5,0) -- (6,0);
\end{tikzpicture}
```


Drawing a thick blue line

```
\begin{tikzpicture}
    \draw[line width=2pt, blue] (0,0) -- (1,0);

    \draw[line width=2pt, red, dotted] (2,0) -- (3,0);

    \draw[line width=2pt, dashed, green] (4,0) -- (5,0);

    \draw [thick,dash dot] (0,1) -- (5,1);
    \draw [thick,dash pattern={on 7pt off 2pt on 1pt off 3pt}] (0,2) -- (5,2);
\end{tikzpicture}
```

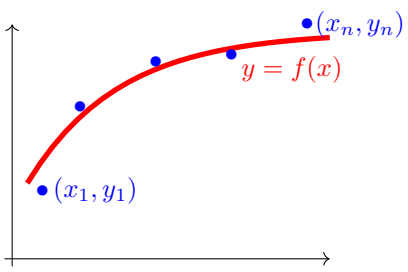
Drawing an arc

```
\begin{tikzpicture}
  \draw [blue,thick,domain=0:270] plot ({5+2.5*cos(\x)}, {1+2.5*sin(\x)});
\end{tikzpicture}
```



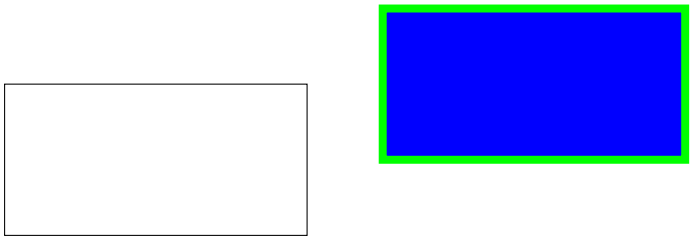
Draw a function

```
\begin{tikzpicture}[
    declare function={%
        F(\x)                    =3-2*pow(2.7979,-0.8*\x);
    }
 ]


    % Zeichnen der Funktion
    \draw[red,line width=2pt,domain=0:4] plot({\x},{F(\x)});

    % Bezeichnung
    \node[red] (O) at(3.5,2.5) {$y=f(x)$};

    % Punkte
    \node[blue] (P1n) at (0.9,0.9) {$(x_1,y_1)$};
    \node[blue] (P1) at (0.2,0.9) {$\bullet$};

   \node[blue] (P2) at (2.7,2.7) {$\bullet$};
   \node[blue] (P3) at (1.7,2.6) {$\bullet$};
   \node[blue] (P4) at (0.7,2) {$\bullet$};

   \node[blue] (P5n) at (4.4,3.1) {$(x_n,y_n)$};
   \node[blue] (P5) at (3.7,3.1) {$\bullet$};

   % Koordinatensystem
   \draw[color=black,->] (-0.2,-0.1) -- (-0.2,3.1);
   \draw[color=black,->] (-0.3,0) -- (4,0);
\end{tikzpicture}
```



Drawing rectangles and moving objects

```
\begin{tikzpicture}
    \draw (0,0) -- (4,0) -- (4,2) -- (0,2) -- cycle;

    \begin{scope}[shift={(5,1)}]
      \draw[green, fill=blue, line width=3pt] (0,0) -- (4,0) -- (4,2) -- (0,2) -- cycle;
    \end{scope}
\end{tikzpicture}
```

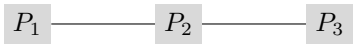


Use of variables

```
\begin{tikzpicture}
\pgfmathsetmacro{\PHI}{-15}
% Now use \PHI anywhere you want -15 to appear,
% can also be used in calculations like 2*\PHI
\def\x{10};

\draw[red] (0,4) -- (1-\PHI*0.5,4);
\draw[green] (0,2) -- (1+1/\x,2);
\draw[blue] (0,0) -- ({1+3*(\x/5+1)},0);
\end{tikzpicture}

\bigskip

%\usetikzlibrary{math} %needed tikz library

\begin{tikzpicture}
  \pgfmathsetmacro{\drehpunktx}{11.63815573}
  %Variables must be declared in a tikzmath environment but
  % can be used outside
  \tikzmath{\x1 = 1; \y1 =1;
  %computations are also possible
  \x2 = \x1 + 1; \y2 =\y1 +3; }
  \draw[->] (\x1, \y1)--(\x2, \y2);
\end{tikzpicture}
```

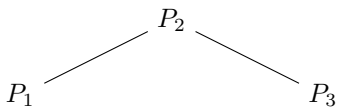
Use of points

```
%\usetikzlibrary{backgrounds}  % is needed

  \begin{tikzpicture}
    \node [fill=gray!30] (P1) at (0,0) { $P_1$ };
    \node [fill=gray!30] (P2) at (2,0) { $P_2$ };
    \node [fill=gray!30] (P3) at (4,0) { $P_3$ };

    \begin{scope}[on background layer]
        \draw (P1) -- (P3);
    \end{scope}
  \end{tikzpicture}
```



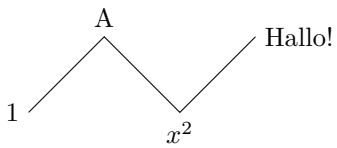
Use of nodes

```
  \begin{tikzpicture}
    \node  (P1) at (0,0){$P_1$};
    \node  (P2) at (2,1){$P_2$};
    \node  (P3) at (4,0){$P_3$};

    \begin{scope}
        \draw (P1) -- (P2) -- (P3);
    \end{scope}
\end{tikzpicture}
```



Use of nodes

```
\begin{tikzpicture}
  \draw (0, 0) node[left]{1}
        -- ++(1, 1) node[above] {A}
        -- ++(1,-1) node[below] {$x^2$}
        -- ++(1, 1) node[right] {Hallo!};
\end{tikzpicture}
```



B. Criteria for a good LaTeX project

A good report is not only characterised by good content, but also fulfils formal aspects. The following list should help to comply with basic rules. Before submitting, all points should be checked and ticked off.

- ☐ Is a suitable directory structure used?
- ☐ Is an appropriate division into files used?
- ☐ Are meaningful directory and file names used?
  - ☐ Are directory and file names such as report or term paper avoided?
  - ☐ Are meaningful names used for images and not „image1“?
  - ☐ Are directory and file names not too long?
  - ☐ Are special characters and spaces avoided?
- ☐ Are commands like \newline and \\ avoided?
- ☐ Are the LaTeXfiles clearly laid out?
  - ☐ Are indentations used in the LaTeXfiles?
  - ☐ Are free lines inserted?
  - ☐ Is the project mentioned in the header of the files?
  - ☐ Does the header of the files mention the main sources?
  - ☐ Is the author mentioned in the header of the files?
- ☐ Are all necessary files given?
- ☐ Are the temporary files deleted?
- ☐ Are graphics created by yourself?
- ☐ Are the sources of the images given?
- ☐ Are citations made with the command \cite?
- ☐ Is a bib file used?
- ☐ Are the entries of the bib-file clearly arranged?
- ☐ Are the entries of the bib-file edited to show them correctly?
- ☐ Are the correct types used for the bib entries?
- ☐ Are meaningful keys used in the bib files?

Unfortunately, the list cannot be complete, but it provides some clues.

C. Model Card

- Model Details - Basic information about the model
  - Person or organization developing model
  - Model date
  - Model version
  - Model type
  - Information about training algorithms, parameters, fairness constraints or other applied approaches, and features
  - Paper or other resource for more information
  - Citation details
  - License
  - Where to send questions or comments about the model
- Intended Use. Use cases that were envisioned during development.
  - Primary intended uses
  - Out-of-scope use cases
- Factors: Factors could include demographic or phenotypic groups, environmental conditions, technical attributes, or others
  - Relevant factors
  - Evaluation factors
- Metrics: Metrics should be chosen to reflect potential real world impacts of the model.
  - Model performance measures
  - Decision thresholds
  - Variation approaches
- Evaluation Data: Details on the dataset(s) used for the quantitative analyses in the card.
  - Datasets
  - Motivation
  - Preprocessing
- Training Data: May not be possible to provide in practice
  - When possible, this section should mirror Evaluation Data. If such detail is not possible, minimal allowable information should be provided here, such as details of the distribution over various factors in the training datasets.
- Quantitative Analyses
  - Unitary results
  - Intersectional results
- Ethical Considerations
- Caveats and Recommendation

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